

Delineating the Upper Bronx River Watershed: Terrain Analysis, Water Quality, and Land Use in an Urban Catchment

By Fiona Dubay

Introduction/Background

A watershed is a hydrologic unit that refers to the entire uphill area of land that drains to one outlet location. Watersheds can be defined on many scales, and are also referred to as catchment areas or drainage basins. Watershed delineation is the process of defining the boundary of a catchment area of interest using topographic data. Defining the boundaries of watersheds is fundamental to understanding the movement, distribution, quantity, and quality of water in a given area. Watershed delineation is of great importance when planning and implementing water-related protection, as it identifies important surface features and clarifies downstream impacts (Sit et al. 2019). Accurate delineation yields valuable information about stormwater management, water quality, water access, the effects of land cover and use, as well as the movement of pollutants through water networks.

Hydrological modeling in urban areas presents some unique challenges. Traditionally, watershed delineation was performed manually with stream network and elevation maps. It wasn't until the 1980s that Digital Elevation Models (DEMs) became widely available (Rumman et al. 2005). Urban hydrology is challenging to model because urban topography is incredibly non-homogenous and non-static (Zhu 2010). The urban landscape is dynamic and ever-expanding, constantly changing the spatiotemporal processes of the water within. It is this unceasing uncertainty in elevation data that makes urban watershed delineation so challenging.

In this paper, watershed delineation in an urban landscape is performed in an attempt to connect water quality to land use. For more than ten years, Riverkeeper has been collecting citizen sampling data on water quality in the Hudson River watershed. Community scientists have collected fecal indicator bacteria from the tributaries and local waterfront areas of the Hudson River. Several of Riverkeeper's sampling sites lay within the Upper Bronx River watershed, and the data from those sites is investigated in this paper.

Research Question

What is the relationship between upslope land cover/use and surface water enterococcus levels in the upper Bronx River watershed? Specifically, this study explores the relationship between the percent impervious surface land cover and surface water enterococcus levels,

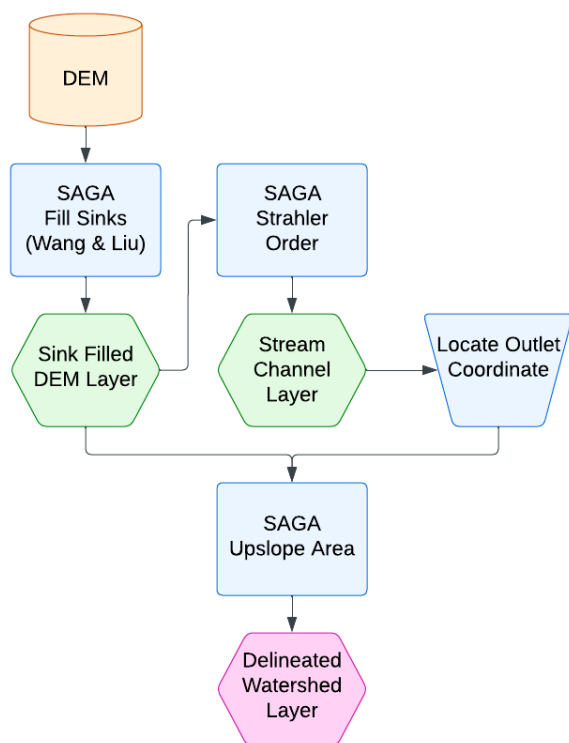
as well as the relationship between the percent developed and percent forested land cover and surface water enterococcus levels in the Upper Bronx River watershed.

Hypothesis

There will be a positive relationship between surface water enterococcus levels in the Bronx River and the percent impervious land cover in the area that drains to the sampling location because fecal indicator bacteria (like enterococci) in overland runoff on impervious surfaces cannot sorb to soil and sediments as it does on pervious surfaces, and will be carried by greater overland flow.

Methodology

Fig.1 - Watershed delineation workflow



The approach used in this study is GIS-based surface hydrological modeling. The DEM used was the USGS National Digital Elevation Model - $\frac{1}{3}$ Arc Second Resolution. The System for Automated Geoscientific Analyses (SAGA) extension for QGIS was used in several different steps. Sinks were filled in the DEM using SAGA's Fill sinks (Wang & Liu) processing tool, which created a new DEM layer (see Figure 1). Using this new filled-in DEM, SAGA's Strahler order tool was used to create a stream channel layer. From there, an outlet point was chosen and used to generate the delineated watershed layer using SAGA's Upslope area tool. Sampling sites from Riverkeeper's dataset within the delineated watershed were chosen. Site coordinates were slightly corrected in order to align with the delineated stream network. For each sampling site, the upslope area was

calculated using SAGA's Upslope area tool, the site's point coordinate, and the sink-filled DEM layer. Using the Multi-Resolution Land Characteristics (MRLC) Consortium's National Land Cover Database (NLCD), land cover and use analyses were made for each site's upslope area. Those analyses include percent developed area, percent forested area, and percent impervious area. The geometric mean of enterococcus counts from 2021 was calculated for each of the chosen sampling sites.

Results

Fig.2 - Stream network and sampling site map

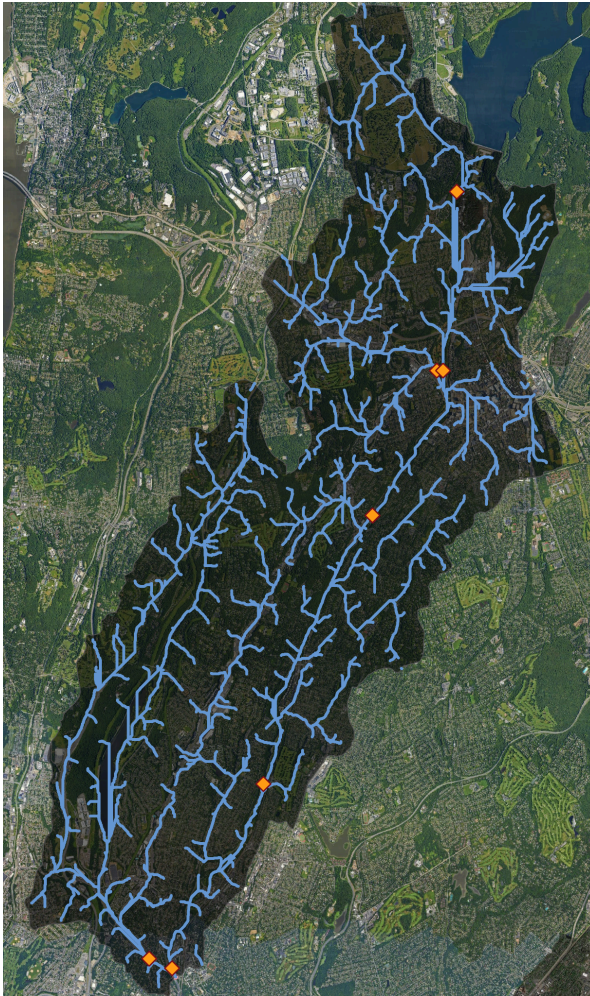


Figure 2 (left) shows the delineated stream channel network as well as the chosen Riverkeeper sampling sites. In order from North to South, the sites are as follows:

- Mount Pleasant S. Kensico Ave at Pat Henry Field
- Westchester County Center
- Greenacres Avenue
- Bronx River Parkway at Leewood Drive
- Sprain Brook at Palmer and Millard Avenue
- Grassy Sprain Brook Confluence

*Fig.3 - Upslope area land use map for
Bronx River Parkway at Leewood Drive*

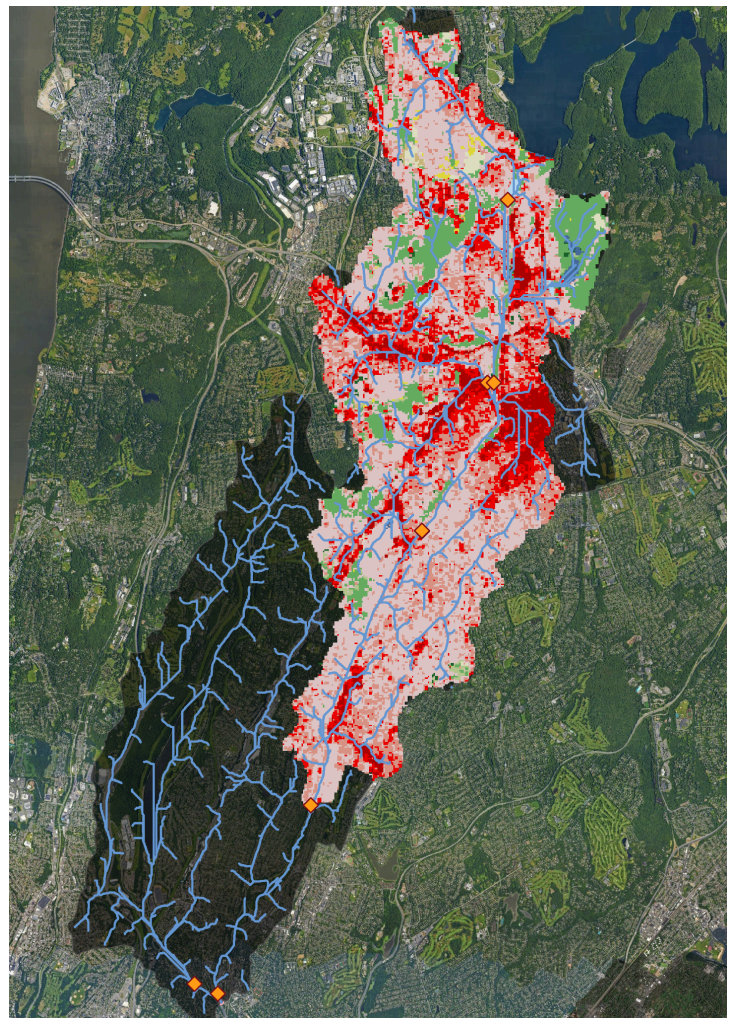


Figure 3 (right) is an example of the upslope area land use maps made for each sampling location. Results from land use analyses can be found in Figure 4, on the following page.

Fig.4 - Upslope area analysis results for sampling sites

Location	Enteroc. count *	Upslope area (mi ²)	% Impervious	% Forested	% Developed
Greenacres Avenue	2022.33	10.85	28.88%	14.52%	82.60%
Bronx River Parkway at Leewood Drive	1111.53	16.34	27.29%	11.26%	86.55%
Mount Pleasant S. Kensico Avenue	2095.12	2.23	20.74%	9.33%	83.54%
Grassy Sprain Brook Confluence	669.97	20.34	28.97%	9.42%	88.68%
Sprain Brook at Palmer and Millard Avenue	2063.82	7.95	26.12%	14.82%	80.62%

Discussion and Conclusions

The relationship between upslope land use and surface stream water quality revealed by this study was much different than expected. There doesn't appear to be any relationship between the two at all. There also doesn't appear to be any relationship between the area of the upslope catchment and the enterococcus counts. This was not expected but could be explained by considering the sewage system in the area. The amount of fecal indicator bacteria could be more dependent on the number of combined sewage overflows (CSOs), other waste leaks, or the differences between sewersheds. Factoring in sewer and septic systems could offer some explanation for the wide range in the geometric means of enterococci counts. Additionally, land use in the Upper Bronx River watershed is more homogenous than expected, with both the percent forested area and percent developed area never varying more than 10% in difference.

The cause of fecal matter contamination of surface streams in the Upper Bronx River watershed is more complicated than the land use in contributing upslope areas. Future work should consider CSO locations and sewersheds, as well as the type of sewage conveyance or treatment methods used within the upslope area of sampling sites.

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